

# Olive Tree Media Limited

## USSD Intergration Documentation

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# USSD Integration Documentation

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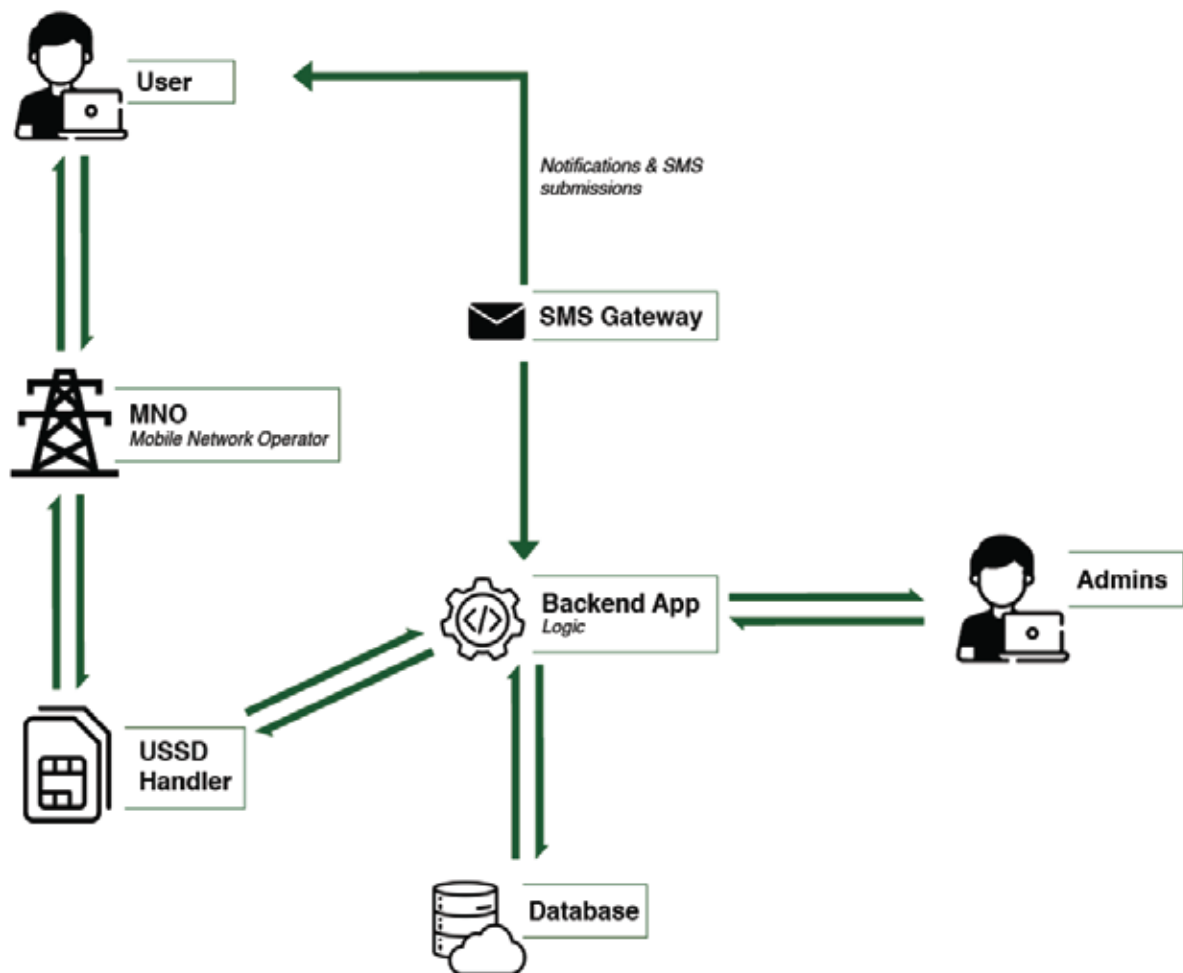
## Overview

USSD (Unstructured Supplementary Service Data) is a session-based communication protocol that enables real-time interaction between mobile devices and applications through short codes. Unlike SMS, USSD maintains an active connection throughout the user interaction, making it ideal for menu-driven services like mobile banking, airtime top-ups, and information services.

## How USSD Works

When a user dials a USSD code (like \*123#), the mobile network operator routes the request to your application via HTTP GET requests. Your application processes the request and responds with either a menu (to continue the session) or a final message (to end the session).

## Overview Architecture of the Project



## API Integration Details

### Request Method

**GET Request** - The Bonga USSD application will make a GET request to your client's exposed URL.

*The Bonga USSD application will make a GET request to the client's exposed URL with the parameters listed when the user dial the code and whenever they respond to a menu flow.*

### Parameters Received

| Parameter     | Description                             | Type   | Example      |
|---------------|-----------------------------------------|--------|--------------|
| mobile_number | Phone Number                            | string | 254712345678 |
| session_id    | Session Number                          | string | 1234567      |
| message       | User Input                              | string | 12500        |
| service_code  | USSD Code                               | string | *123#        |
| network       | Mobile Network Operator of Phone Number | string | Safaricom    |

### Understanding the Message Parameter

The **message** parameter is crucial for understanding user interactions:

- **First dial:** Empty string "" when user initially dials the USSD code
- **Subsequent inputs:** User responses are concatenated with asterisks (\*) as delimiters
- **Example progression:** "" → "1" → "1\*2" → "1\*2\*500"

# Client Application Responses

USSD being a session based technology, when a user dials the USSD code or responds to a menu the provided url will receive a call with the above parameters.

Respond beginning with:

- **CON** - if the USSD session is ongoing e.g. **CON Welcome to our service**
- **END** - if you want to end the USSD session e.g. **END Your request has been received**

Your application must respond with specific prefixes to control the USSD session:

## Continue Response (CON)

User **CON** when you want to keep the session active and show a menu or prompt for more input:

Example:

*CON Welcome to Our Service*  
*1. Check Balance*  
*2. Transfer Money*  
*3. Pay Bills*

## End Response (END)

Use **END** when you want to terminate the USSD session with a final message:

Example:

*END Your request has been received*  
*Transacation ID: TXN12345*  
*Thank you for using our service*

## Sample URL Requests

### Initial Request (User Dials \*123#)

*GET https://yourapp.com/ussd?*  
*mobile\_number=254712345678&session\_id=1234567&message=&service\_code=\*123#&network=Safaricom*

### After User Selects Option 1

*GET https://yourapp.com/ussd?*  
*mobile\_number=254712345678&session\_id=1234567&message=1&service\_code=\*123#&network=Safaricom*

### After User Enters Amount (500)

*GET https://yourapp.com/ussd?*  
*mobile\_number=254712345678&session\_id=1234567&message=1\*500&service\_code=\*123#&network=Safaricom*

# Complete Session Flow Example

Here's a step-by-step example of a money transfer service:

## Step 1: Initial Dial

User Action: Dials \*123# System Receives: *message="" (empty string)*

System Responds:

CON Main Menu  
1. Check Balance  
2. Transfer Money  
3. Pay Bills  
0. Exit

## Step 2: User Selects Transfer Money

User Action: Presses 2 System Receives: *message="2"*

System Responds:

CON Transfer Money  
Enter recipient phone number:

## Step 3: User Enters Phone Number

User Action: Types 254700123456 System Receives: *message="2\*254700123456"*

System Responds:

CON Transfer to 254700123456  
Enter amount (KES):

## Step 4: User Enters Amount

User Action: Types 500 System Receives: *message="2\*254700123456\*500"*

System Responds:

CON Confrim Transfer  
Amount: KES 500  
To: 254700123456  
1. Confrim  
2. Cancel

## Step 5: User Confirms

User Action: Presses 1 System Receives: *message="2\*254700123456\*500\*1"*

System Responds:

END Transfer Successsful  
KES 500 sent to 254700123456  
Transaction ID: TXN789012  
Your new balance: kes 2,500

# Best Practices for Developers

## **Menu Design**

- Keep menus simple and intuitive
- Use numbered options (1, 2, 3...)
- Include a "0" option for going back or exiting
- Limit menu options to 9 or fewer for better user experience

## **Message Handling**

- Always check if **message** is empty to detect initial dial
- Parse the message by splitting on asterisks (\*) to get individual user inputs
- Validate user inputs at each step
- Handle invalid inputs gracefully

## **Session Management**

- Use the **session\_id** to track user sessions
- Store session state if needed (database, cache, etc.)
- Set reasonable session timeouts
- Clean up session data after completion

## **Error Handling**

- Validate phone numbers and amounts
- Provide clear error messages
- Always end sessions with **END** after errors
- Include helpful instructions in error messages

## **User Experience**

- Keep text concise (USSD has character limits)
- Use clear, simple language
- Provide confirmation steps for transactions
- Show transaction receipts or confirmation numbers

# Testing Your Integration

## *Test Cases to Consider*

1. **Initial dial** - Verify empty message handling
2. **Menu navigation** - Test all menu options
3. **Invalid inputs** - Test non-numeric inputs where numbers expected
4. **Session timeout** - Test behavior when session expires
5. **Network variations** - Test with different mobile network operators
6. **Edge cases** - Test with very long phone numbers or amounts

## *Common Issues*

- Not handling empty message on initial dial
- Incorrect response format (missing CON/END prefix)
- Session data not being maintained between requests
- Not validating user inputs properly
- Response text too long for USSD display

## Next Steps

1. **Set up your endpoint** - Create a URL that can receive GET requests
2. **Implement response logic** - Handle the parameters and return appropriate CON/END responses
3. **Test thoroughly** - Use the test cases mentioned above
4. **Deploy and monitor** - Go live and monitor for any issues
5. **Get support** - Contact us if you need assistance



## Support and Contact

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**Support:** [support@olivetreemobile.co](mailto:support@olivetreemobile.co)

**Need Help?** Our technical support team is available to assist with your USSD integration. Whether you're facing issues with session management, response formatting, or testing, we're here to help ensure your USSD service works seamlessly.